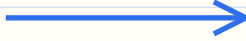
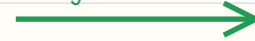


ch.1



lets go!



Modal Testing: A Practitioner's Guide

Ch. 1 - Experimental Modal Analysis without tears

detailed handwritten-style notes - trial edition

Whole chapter in 1 line:

a structure = a *team of SDOF oscillators* glued together. Modal analysis names each player: *frequency + damping + shape*. Everything else = how to see them clearly.

How to use:

Made for a MechE fresher who knows basic vibrations. Start at S0, read in order, do every [TRY]. Finish with the self-test on the last pages!

Legend: [KEY] key idea [LAB] lab tip [TRAP] trap [LINK] to basics [TRY] try it

Council crew: Dr. Meera K. (theory) - Dr. Viktor H. (lab) - Dr. Lena F. (signals) - Prof. Arjun D. (pedagogy) - Dr. Sofia M. (field)

| 11 Sep 2026



(0) Primer - from SDOF to modes

$$m x'' + c x' + k x = f(t) \rightarrow f(n) = (1 / 2 \pi) \times \sqrt{k / m}$$

Your upgrade map:

[LINK] You know *1 mass gives 1 peak*. A plate or beam is many masses, so *many peaks*. Each peak = one 'hidden SDOF' = one *mode*. Resonance in SDOF becomes: every mode resonates *in its own shape*, with *its own damping*. Messy time traces become clear peak plots after the *FFT*.

You know (SDOF)	This chapter adds (real structures)
1 natural frequency	MANY natural frequencies, one per mode
Resonance = big response near f_n	Each mode resonates in its own shape
Damping ratio sets peak height	Every mode has its own damping
FRF of 1 oscillator: 1 peak	FRF of a structure: many peaks
Time response tells the story	Frequency domain (FFT) tells the story

Kitchen modal test:

[TRY] Hum near a steel plate and hear it ring at certain pitches - those are its natural frequencies. Tap different spots to excite them differently. That IS the plate experiment of S1.1, in your kitchen!

(1.1) Could you explain modal analysis to me?

The 3-step engineering logic (cantilever story):

1. Characteristics (length, E, I, density...) describe the beam but alone say nothing about deflection or failure. **2. + Loads** gives response. **3. + Design spec** gives pass/fail. Modal data is step-1 data for dynamics: **powerful, but cannot judge alone!** (In real troubleshooting, loads and specs are often unknown - that is exactly when modal thinking saves you.)

[KEY] Structural dynamics = the full movie: all inputs (fan, disk drive, bumps) give total response. **Modal analysis** = the cast list: the system's own modes. Time traces look chaotic, but in the **frequency domain** the chaos separates into clear peaks sitting at the modes.

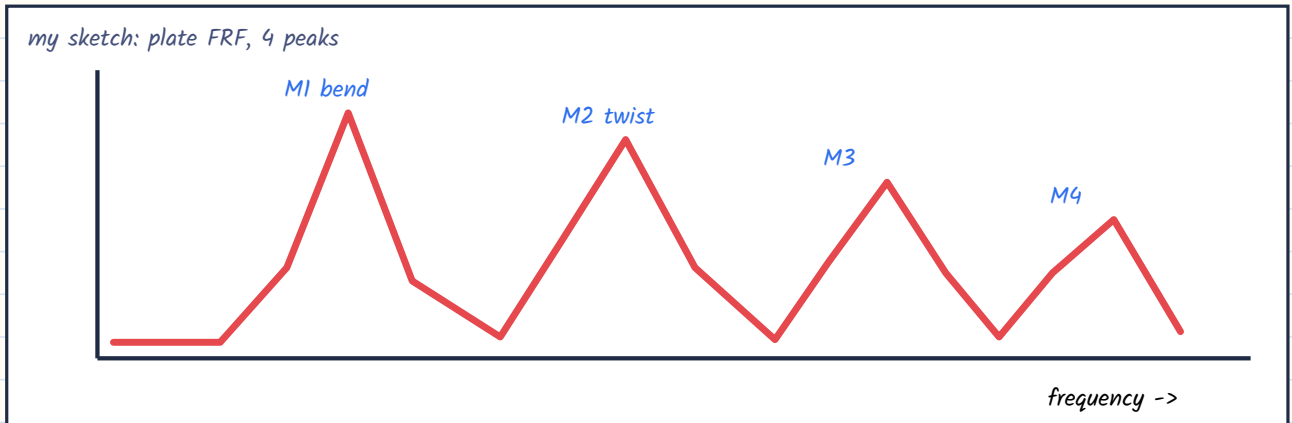
Modes are band-pass filters:

Each mode amplifies input energy near its own frequency and ignores the rest. Measured response = sum over modes of (mode filter x input). Uneven input spectrum gives uneven peak heights - even a strong mode looks small if starved of input energy there!

The plate experiment (heart of S1.1): free plate, sine force of constant peak, varying frequency at one corner, accelerometer at another corner.

-> **Sine sweep in time:** response breathes up and down as excitation frequency slides - same force, wildly different response. Maxima = crossing a **resonance**.

- > **FFT gives the FRF:** 4 clean peaks (this plate, this band). Overlay time-sweep maxima on FRF peaks - they line up. **[TRAP]** A random time trace would be unreadable; the FRF rescues us.
- > **Dwell at each peak** with 45 accelerometers: freeze the excitation at one natural frequency and map deformation everywhere: M1 = 1st bending, M2 = 1st twist, M3 = 2nd bending, M4 = 2nd twist. Strictly these are operating deflections, practically they equal the mode shapes for well-separated modes.



[KEY] Mass and stiffness place the modes; damping sizes the peaks. Designers use modes to avoid resonances; test engineers use them to diagnose noise and vibration failures.

Two analogies - interview gold:

Cookbook: hundreds of ingredients, but each recipe uses its own small subset in its own proportions. Each loading 'cooks' its own subset of modes.

100-piece orchestra: each score uses different instruments at different intensities; one out-of-tune player ruins the piece, and you find the culprit only by listening to players individually. Total response hides the fault; modes let you audition each player separately - modal analysis's superpower!

Council bonus, playground swings: random pushes = weak chaos (broadband input, messy trace); pushing in rhythm = giant motion (dwell at resonance). Two swings tied with a rope = coupled modes, like bend + twist of the plate!

[TRY] Tap a ruler clamped to a table at different overhang lengths - pitch drops as overhang grows (stiffness falls). You just did 'mass and stiffness place the modes' with stationery.

(1.2) Just what are FRFs?

$$H(f) = X(f) / F(f) \text{ ...complex! mag + phase = real + imag}$$

Measure **force AND response together** (response = displacement, velocity, or acceleration), FFT both, divide. All four views (magnitude, phase, real, imaginary) show the same truth.

The 3-DOF beam gives a 3 x 3 FRF matrix:

	force @ 1	force @ 2	force @ 3
resp @ 1	h11	h12	h13
resp @ 2	h21	h22	h23
resp @ 3	h31	h32	h33 <- ref 3 row

Notation $h(\text{out}, \text{in})$: matrix row = response point, column = force point. 3 push points x 3 measure points = 9 FRFs.

Drive-point FRF (push + measure at SAME point, e.g. h_{33}) - memorize like a phone number:

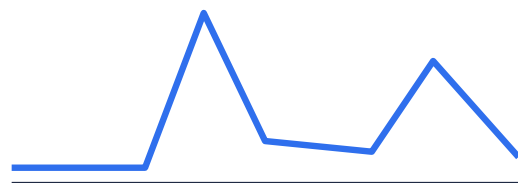
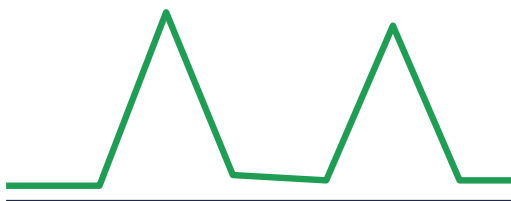
1. Peaks (resonances) and valleys (anti-resonances) **alternate**.
2. Phase **drops 180 deg** over a resonance, **gains 180 deg** over an anti-resonance.
3. All **imaginary-part peaks point the same way**.

[LAB] First check on any drive-point plot: peak-valley alternation? Imag peaks same direction? If not, suspect the measurement, not the theory. - Viktor

drive point vs cross FRF

drive h_{33} : peak-v-peak!

cross: no alternation



Reciprocity - you do NOT need all 9 FRFs:

[KEY] Because the M, C, K matrices are symmetric, $h(ij) = h(ji)$: push at i / measure at j equals push at j / measure at i. So **one row OR one column** of the matrix is enough to get every mode shape. (The book's 15-point waterfall plot shows the same idea: imag-peak ridges trace each shape.)

1.2.1 Why is only one row or column needed?

-> Read the **imaginary-peak heights** across one row and the shape pops out. Row 3 and row 2 both show mode 1; heights differ by a scale factor - the shape is what matters.

THE NODE RULE:

[TRAP] With the reference at point 2, **mode 2 vanishes** - point 2 sits on mode 2's node (zero-motion line). **Never put your reference on (or near) a node of a mode you need!** Later chapters turn this into reference-selection strategy.

[TRY] Micro-example: imag peaks [+0.2, +0.9, +1.6] read first-bending (same sign, growing). Peaks [+1.0, 0.0, -1.0] read second-mode-like with a node in the middle - and warn you a reference at the middle would miss this mode entirely.

(1.3) Shaker test vs impact test?

	Hammer (impact)	Shaker
You get	One ROW (response ref fixed, hammer roves)	One COLUMN (force ref fixed, accels rove)
Theory	identical!	identical! ($h_{ij} = h_{ji}$)

	<i>Hammer (impact)</i>	<i>Shaker</i>
Best at	Fast, portable, no attached hardware	Controlled input; big / heavy / damped parts
Main enemies	Tip choice, double hits, leakage	Mass loading, stinger effects

The test article is bigger than you think:

[KEY] Test article = structure + *everything touching it*: suspension, cables, sensor masses, shaker + stinger. Theory assumes massless sensors and perfect forces - the lab does not.

1. **Roving-mass effect**: one accelerometer is nothing vs the whole machine but huge vs a thin panel's local mass. Moving sensors between readings retunes the structure mid-test. **[LAB]** Fix: leave ALL accelerometers mounted (use a few at a time), or add **dummy masses** at unmeasured points.

2. **Shaker / stinger**: the stinger should push purely axially and decouple shaker dynamics, but residual mass and stiffness effects often survive. Impact tests dodge this entirely - hence hammer-vs-shaker differences on the same part.

1.3.1 What do we measure to compute the FRF? (analyzer chain)

sensor -> anti-alias -> ADC -> window -> FFT -> avg spectra -> FRF + coherence

<i>Stage</i>	<i>What happens</i>	<i>Enemy</i>
Transducers	Force cell + accel / vel / disp pickup	Bad mounting, saturation
ADC (12/16/24-bit)	Digitizes the waveform	Sampling (resolution) + quantization (amplitude steps)
Window	Weights record so FFT periodicity is nearly met	Leakage if skipped wrongly
FFT + averaged spectra	Input/output auto-spectra + cross-spectrum	Noise (averaging fights it)
FRF + coherence	FRF = cross/input (essentially); coherence 0-1	Coherence well under 1: do not trust band

LEAKAGE - villain no. 1:

[KEY] The FFT assumes your record covers all time or *repeats forever*. A cut-off sine violates that, so energy leaks and smears across frequencies. Fix = **windows**: weighting functions that taper the record so it looks periodic. Windows distort a little; leakage distorts catastrophically. (S1.4 to S1.6 = the war against leakage.)

Lena's one-liner: sample rate picks your map's borders, bits pick its resolution, windows pick your poison - leakage or taper. Choose taper.

(1.4 + 1.5) Impact and shaker essentials

Impact testing: two headline items (full war stories in Ch.7):



Hammer-tip hardness = your frequency-range knob:

[LAB] Harder tip gives wider excited band (short sharp pulse); softer tip gives narrower band (long gentle push). General rule with exceptions - verify on screen. Good = flat-ish input spectrum + coherence near 1 across the band. Bad = spectrum rolling off + coherence and FRF dying in the upper half (you did not feed those modes energy).

[TRAP] A soft tip is not 'wrong' - only wrong if you needed the higher modes. If your band of interest is the flat part, ship it!

The response must die before the record ends:

Lightly damped structures keep ringing past the sample interval, which means leakage. Standard rescue = **exponential window** on the response (decays the tail toward zero).

[KEY] Windows damage data - try the cures first: 1. **narrower frequency span** (longer time record), 2. **more spectral lines** (finer resolution, longer record). Both give the vibration more time to decay naturally.

[LAB] Viktor's pre-flight: tip? decay? coherence? - three glances before I believe any hammer FRF. Plus: no **double hits**, hit squarely, watch for filter ring and saturated sensors (Ch.7 previews).

Shaker testing: pick an excitation that needs NO window:

Excitation	Leakage?	Window?	Notes
Burst random	NO - transient + decay fit ONE record	None	Workhorse of modal testing
Sine chirp	NO - repeats exactly in record	None	Great control of the band
Plain random	YES - never periodic in window	Hanning (must); still a bit distorted	Easy to run, weakest data

Memory hook:

Lena: 'Random + Hanning is the fast food of excitation: convenient, satisfying, slightly guilty. Burst random and chirp are home cooking.'

Burst and Chirp = Bye-Ciao windows! (Burst fits the Box; Chirp Cycles exactly.)

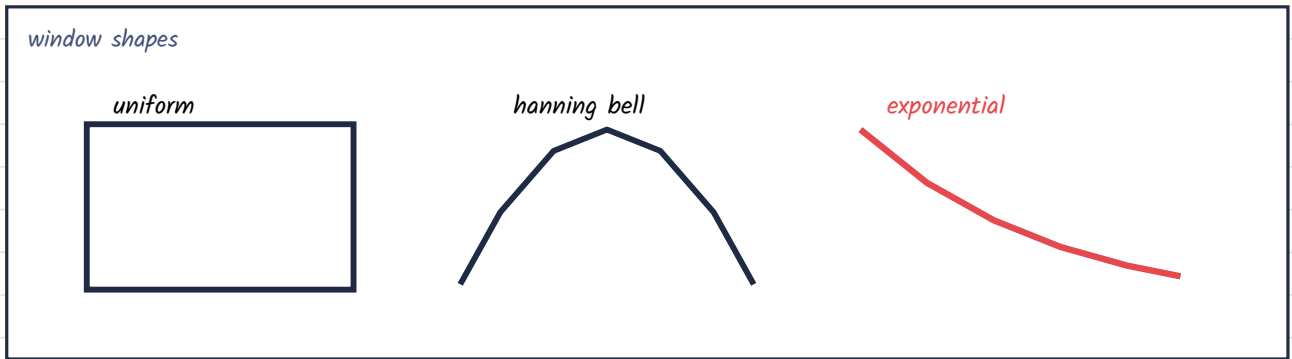
(1.6) Tell me more about windows!

Motto: no window if you can avoid it; the right window if you can't (field testing and operating data often give no choice).

Window	Shape in words	Use it when...
Uniform (rect / boxcar / 'no window')	Gain = 1 everywhere	Signal fully in one record OR periodic-in-window
Hanning	Bell / cosine, ends forced to 0	Random / field signals that break periodicity
Flat-top	Broad flat crown	Pure sine of unknown period; amplitude accuracy matters

Window	Shape in words	Use it when...
Force + Exponential	1 around hammer pulse; decaying curve	Impact pair: isolate pulse + decay ringing tail

Typical pairings: uniform with impact (pulse + decay fit), burst random, chirp, pseudo-random, stepped sine; Hanning with shaker random and operating data; flat-top with calibration and constant-speed (RPM) excitation.



Price of every window:

[TRAP] Peak amplitudes get less accurate and the structure *looks more damped than it is*. Acceptable - leakage's smearing is far worse. Deep math lives in the signal-processing chapter.

[TRY] One-question drill: burst random test - which window? Uniform / none. Factory-floor random? Hanning. Hammer test, response ringing past the record? Force + exponential. Sine calibration? Flat-top.

(1.7) How do we get mode shapes from plate FRFs?

Step 1 - peak-picking (the simple, honest start):

6 points give $6 \times 6 = 36$ possible FRFs. Read the *imaginary-part peaks* mode by mode:

- > **Mode 1 (bending)**: points 1, 2, 5, 6 near -1; points 3, 4 near +1. Classic first-bending arc (all 45 points confirm it).
- > **Mode 2 (torsion)**: 1:+2, 2:-2, 5:-2, 6:+2, and 3, 4 near 0. Twist with a *node line* through 3-4. Hello again, node rule from S1.2!

Step 2 - curvefitting (what software really does):

Peak-picking is fine for simple, well-separated modes. Production work uses *modal parameter estimation* = **curvefitting**: decompose the measured FRF into a *sum of SDOF oscillators* (the S0 mental model becomes literal):

$$\text{measured FRF} = \text{SDOF}(M1) + \text{SDOF}(M2) + \dots + \text{residuals}$$

The analyst's three inputs (your job; the algorithm does the rest):

1. **Frequency band** to fit (fit in bands, a few modes at a time).
2. **How many modes** live in that band (hard when modes crowd - indicator tools help; full toolbox in Ch.5 / Ch.9).
3. **Residual terms** - compensation for modes outside the band still tugging on it.

[TRAP] Garbage in, garbage out: curvefitting cannot rescue bad FRFs – which is why S1.3-S1.6 (measurement quality) come before estimation in a practitioner's head. Sofia: 'I have never seen a curvefitter fix a double-hit. I have seen many analysts try.'

(1.8) Modal data vs operating data – do not mix!

1.8.1 What is operating data?

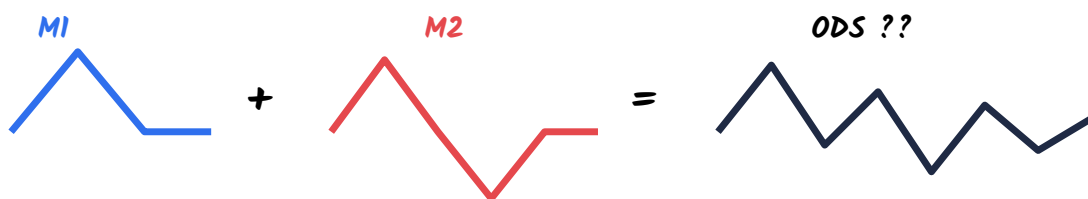
Operating data = response ONLY.

[KEY] The machine runs (forces unknown / unmeasured) and you record vibration. Golden chain: **response = FRF (system filter) x force (whatever it is)**. The FRF and modes decide how the structure wants to move; the operating force decides which modes play and how loudly.

Plate demo with a single-sine force at one corner (2 modes for simplicity):

- > **Excite near mode 1:** operating deflection shape looks like **mode 1** (+ a whisper of mode 2).
- > **Excite near mode 2:** shape looks like **mode 2** (+ a whisper of mode 1).
- > **Excite midway:** a strange hybrid nobody recognizes – until you decompose it: a little bending + a little torsion. **Operating shapes are linear combinations of mode shapes.** With broadband force, many modes join the sum.

operating shape = mode 1 + mode 2 mixed!



[TRAP] With operating data you never measure the force or the FRF – so you cannot see why the shape looks that way. That is the whole argument for modal testing.

1.8.2 So what good is modal data? (payoff slide)

Superpower	What it means	Needs
SDM (structural dynamic modification)	Predict 'what if we add ribs / mass / damping HERE?' without cutting metal	Modal model (f, damping, shapes)
Forced-response simulation	Predict response to ANY hypothetical force	Modal model
FEM correlation + updating	Check and fix the finite-element model against reality	Modal model + FEM

[KEY] Modal data = the reusable model of the system. Operating data = a snapshot of one situation.

1.8.3 Should I collect modal or operating data?

Both, whenever schedule and budget allow. If forced to choose, know what you lose:

	Modal data	Operating data
Measures	Force AND response -> FRF -> true f, damping, shapes	Response only
Gives	System characteristics; SDM + simulation + FEM work	True in-service behavior
Cannot	Judge pass/fail alone (needs loads + spec - S1.1!)	Do SDM / sim; shapes often confuse

Sofia's field rule + mini-case:

'Operating data tells you *where it hurts*; modal data tells you *why, and what to change*. Painkillers or cure - your call.'

Mini-case: a car cabin drones at 2800 RPM. Operating test shows a hot floor panel - but which fix: stiffen, damp, or move the exhaust hanger? Modal test reveals a floor bending mode exactly at that RPM's firing frequency with a belly at the hot spot - targeted rib + damping patch. Operating found the pain; modal prescribed the cure.



(1.9 + Revision) Closing + one-page recap

Ch.1 gave the **non-mathematical intuition**: modes as filters, FRFs as the measurement, hammer vs shaker, leakage vs windows, peak-pick to curvefit, modal vs operating. Next: **Ch.2** - the math under the intuition (SDOF to MDOF theory); **Ch.3-5** - signal processing, excitation details, parameter estimation; **Ch.6+** - the practitioner's craft.

[LINK] Your study loop from here: for every later equation, ask 'which Ch.1 picture does this formalize?' - if you can answer, you own the chapter.

#	Big idea	Punchline
1	Modal analysis finds...	f, damping, shape per mode: the dynamic fingerprint
2	...but cannot judge alone	Needs loads + spec (cantilever lesson)
3	FRF	Output / input in freq domain; complex
4	Drive point hii	Alternate peak/valley; -/+180 deg flips; imag same side
5	Reciprocity	hij = hji: one row OR column suffices
6	Node rule	Reference on a node means mode invisible
7	Shaker vs hammer	Theory same; practice differs (mass, stinger)
8	FRF chain	Sense -> AA -> ADC -> window -> FFT -> FRF + coh
9	Leakage	FFT demands periodicity; windows are the tax
10	Hammer tip = band knob	Hard = wide; flat spectrum + coh near 1 = good
11	Ringling tail?	Exponential window - or longer record
12	Best shaker inputs	Burst random + sine chirp = window-free
13	Windows	Uniform / Hanning / flat-top / force+exp
14	Curvefitting	FRF = sum of SDOFs + residuals
15	Operating vs modal	ODS = sum modes x force; modal = reusable model

$f_n = (1/2\pi) \sqrt{k/m}$ - $H = X/F$ - $h_{ij} = h_{ji}$ - $resp = FRF \times force$ - $coh \text{ near } 1 = \text{trust}$



(Self-test) 10 questions – no peeking!

- * **Q1.** Why can't modal data alone say a design is acceptable?
- * **Q2.** Same peak force, swept frequency, yet response varies wildly. Why?
- * **Q3.** FRF shows 4 peaks but the time trace is chaos. Which do you trust for natural frequencies, and why?
- * **Q4.** Name the 3 drive-point signatures.
- * **Q5.** Your reference accelerometer sits on a mode's node. What happens? Fix?
- * **Q6.** Roving hammer gives a ____ of the FRF matrix; shaker gives a ____.
- * **Q7.** Hammer spectrum rolls off and coherence dies above 800 Hz. Diagnose + fix.
- * **Q8.** Which two shaker excitations need no window, and why is each leakage-free?
- * **Q9.** Burst random / factory random / ringing hammer response / sine calibration: pick windows.
- * **Q10.** ODS at 55 Hz looks like nothing seen before; modes sit at 50 and 60 Hz. Explain.

Answers (really no peeking!):

- A1.** Modes are characteristics only; response needs loads, verdict needs spec (S1.1).
- A2.** You are crossing resonances – each mode amplifies input near its own frequency (modal filtering).
- A3.** The FRF – the FFT separates the modal filters; time superposition hides them.
- A4.** Peak/valley alternation; -180 deg at resonance / $+180$ at anti-resonance; imag peaks all same direction.
- A5.** That mode vanishes from every FRF. Move the reference off the node.
- A6.** Row; column (fixed response ref vs fixed force ref).
- A7.** Tip too soft – no energy up high. Use a harder tip (verify flat spectrum + coh near 1); if only under 800 Hz matters, it is fine.
- A8.** Burst random (whole transient + decay in one record) and sine chirp (exactly periodic in the record).
- A9.** Uniform / none; Hanning; force + exponential; flat-top.
- A10.** 55 Hz sits between modes, so ODS = combination of both (mostly), not a pure mode shape.

Tear-off lab checklist:

Before: band of interest? tip / excitation picked for it? reference off all nodes? all sensors mounted (no roving-mass surprise)? stinger aligned (shaker)?

During: no double hits, square impacts? response decayed in-record (or windowed)? no overload / saturation? coherence near 1 in band?

After: drive-point checks pass? reciprocity spot-check (hij near hji)? logbook filled (tip, span, lines, windows, averages)?

(Glossary) First-use definitions, Ch.1

Mode - one natural pattern: frequency + damping + shape.

Natural frequency - where the mode amplifies most.

Mode shape - the pattern: relative amplitudes / phases.

Damping - energy loss per cycle; sets peak width / height.

FRF - complex output / input vs frequency.

Drive point - FRF with force and response at the same DOF.

Cross FRF - force and response at different DOFs.

Anti-resonance - valley between peaks (drive-point).

Reciprocity - $h_{ij} = h_{ji}$.

Node - zero-motion line / point of a mode.

Reference - the fixed DOF of a row / column survey.

FFT - time-to-frequency transform.

Leakage - smearing when the record is not periodic-in-window.

Window - taper weighting that fakes periodicity.

Aliasing - high-frequency fold-back from bad sampling / filtering.

ADC / quantization - digitizer; bit depth sets amplitude resolution.

Auto / cross spectrum - averaged building blocks of FRF / coherence.

Coherence - 0-to-1 data-quality meter.

Burst random / sine chirp / random - shaker signals (S1.5).

Peak-picking - reading shapes from imag-peak heights.

Curvefitting - decomposing FRFs into SDOF contributions (+ residuals).

Residual - out-of-band modes' leftover influence.

ODS - operating deflection shape: response-only shape at one frequency.

SDM - predicting design changes from the modal model.

FEM correlation - validating / updating simulation with modal test data.

Council sign-off: Meera (theory) ok - Viktor (lab) ok - Lena (signals) ok - Arjun (pedagogy) ok - Sofia (field) ok.

Notes are original summaries; approximations labelled. Next stop: Ch.2, the math!

